

7.查看和调整CPU、NPU和DDR频率

1. 查看和固定CPU频率

查看 CPU 当前运行频率

代码块

```
1 cat /sys/devices/system/cpu/cpu0/cpufreq/scaling_cur_freq
2 cat /sys/devices/system/cpu/cpu1/cpufreq/scaling_cur_freq
3 cat /sys/devices/system/cpu/cpu2/cpufreq/scaling_cur_freq
4 cat /sys/devices/system/cpu/cpu3/cpufreq/scaling_cur_freq
5 cat /sys/devices/system/cpu/cpu4/cpufreq/scaling_cur_freq
6 cat /sys/devices/system/cpu/cpu5/cpufreq/scaling_cur_freq
7 cat /sys/devices/system/cpu/cpu6/cpufreq/scaling_cur_freq
8 cat /sys/devices/system/cpu/cpu7/cpufreq/scaling_cur_freq
```

查看 CPU 可用频率

在串口终端中执行以下命令查看 CPU 可用的频率。

通过以下命令查看 CPU 的频率，其中 policy0 代表可以控制查看 CPU0 到 CPU3 的频率，policy4代表可以控制和查看 CPU4 和CPU5 的频率，policy6 代表可以控制和查看 CPU6 和CPU7的频率

代码块

```
1 # 查看每个 policy 管理哪些 CPU
2 cat /sys/devices/system/cpu/cpufreq/policy0/affected_cpus # 输出: 0 1 2 3
3 cat /sys/devices/system/cpu/cpufreq/policy4/affected_cpus # 输出: 4 5
4 cat /sys/devices/system/cpu/cpufreq/policy6/affected_cpus # 输出: 6 7
5
6 # 查看可用的频率
7 cat /sys/devices/system/cpu/cpufreq/policy0/scaling_available_frequencies
8 cat /sys/devices/system/cpu/cpufreq/policy4/scaling_available_frequencies
9 cat /sys/devices/system/cpu/cpufreq/policy6/scaling_available_frequencies
```

```
root@topeet:/usr/bin# cat /sys/devices/system/cpu/cpufreq/policy0/scaling_available_frequencies
408000 600000 816000 1008000 1200000 1416000 1608000 1800000
root@topeet:/usr/bin# cat /sys/devices/system/cpu/cpufreq/policy4/scaling_available_frequencies
408000 600000 816000 1008000 1200000 1416000 1608000 1800000 2016000 2208000 2400000
root@topeet:/usr/bin# cat /sys/devices/system/cpu/cpufreq/policy6/scaling_available_frequencies
408000 600000 816000 1008000 1200000 1416000 1608000 1800000 2016000 2208000 2400000
root@topeet:/usr/bin# █
```

设置 CPU 频率为性能模式

可以通过以下命令将 CPU 设置为性能模式。

代码块

```
1 echo performance > /sys/devices/system/cpu/cpufreq/policy0/scaling_governor
2 echo performance > /sys/devices/system/cpu/cpufreq/policy4/scaling_governor
3 echo performance > /sys/devices/system/cpu/cpufreq/policy6/scaling_governor
```

设置完成在 rk3588 终端上再执行以下命令查看一下当前状态下每个 CPU 频率。

代码块

```
1 cat /sys/devices/system/cpu/cpu0/cpufreq/scaling_cur_freq
2 cat /sys/devices/system/cpu/cpu1/cpufreq/scaling_cur_freq
3 cat /sys/devices/system/cpu/cpu2/cpufreq/scaling_cur_freq
4 cat /sys/devices/system/cpu/cpu3/cpufreq/scaling_cur_freq
5 cat /sys/devices/system/cpu/cpu4/cpufreq/scaling_cur_freq
6 cat /sys/devices/system/cpu/cpu5/cpufreq/scaling_cur_freq
7 cat /sys/devices/system/cpu/cpu6/cpufreq/scaling_cur_freq
8 cat /sys/devices/system/cpu/cpu7/cpufreq/scaling_cur_freq
```

```
root@topeet:/usr/bin# cat /sys/devices/system/cpu/cpu*/cpufreq/scaling_cur_freq
1800000
1800000
1800000
1800000
2400000
2400000
2400000
2400000
root@topeet:/usr/bin#
```

2. 查看和固定NPU频率

查看 NPU 频率

在 rk3588 上查看当前 NPU 频率。

代码块

```
1 cat /sys/class/devfreq/fdab0000.npu/cur_freq
```

查看 NPU 可用频率

在 rk3588 上查看 NPU 可用频率。

代码块

```
1 cat /sys/class/devfreq/fdab0000.npu/available_frequencies
```

设置 NPU 频率

在 rk3588 开发板上，设置 npu 频率之前需要先将给电源开启。

代码块

```
1 echo userspace > /sys/class/devfreq/fdab0000.npu/governor
2
3 echo 6000000000 > /sys/class/devfreq/fdab0000.npu/userspace/set_freq
4 echo 10000000000 > /sys/class/devfreq/fdab0000.npu/userspace/set_freq
```

设置完成可以查看 npu 当前频率。

代码块

```
1 cat /sys/class/devfreq/fdab0000.npu/cur_freq
```

3. 查看 NPU 占用率

在 NPU 运行过程中，可以使用以下命令去查看 NPU 当前占用率，rk3588 有三个 npu 核心，会打印三个 npu 的使用率。

代码块

```
1 cat /sys/kernel/debug/rknpu/load
```

```
root@topeet:/usr/bin# cat /sys/kernel/debug/rknpu/load
NPU load:  Core0:  0%, Core1:  0%, Core2:  0%,
root@topeet:/usr/bin#
```

4. DDR 频率

查看当前 DDR 实际运行频率

代码块

```
1 cat /sys/kernel/debug/clk/clk_summary | grep ddr
```

找到 scmi_clk_ddr 这一行，后面的数字就是当前 DDR 的实时频率（单位是赫兹）

```
root@topeet:/userdata# cat /sys/kernel/debug/clk/clk_summary | grep ddr
scmi_clk_ddr          0      0      0 1596000000      0      0 50000
clk_ddr_fail_safe    0      0      0 240000000      0      0 50000
tclk_wdt_ddr         0      0      0 240000000      0      0 50000
clk_ddr_timer_root   0      0      0 240000000      0      0 50000
clk_ddr_timer1       0      0      0 240000000      0      0 50000
clk_ddr_timer0       0      0      0 240000000      0      0 50000
fclk_ddr_cm0_core    1      1      0 396000000      0      0 50000
aclk_ddr_sharemem    1      1      0 500000000      0      0 50000
pclk_dma2ddr         1      1      0 100000000      0      0 50000
pclk_ddrcm0_intmux   1      1      0 100000000      0      0 50000
clk_ddr_cm0_rtc      1      1      0 32768          0      0 50000
aclk_dma2ddr         1      1      0 702000000      0      0 50000
root@topeet:/userdata#
root@topeet:/userdata#
root@topeet:/userdata#
```

查看可设置的频率

代码块

```
1 cat /sys/class/devfreq/dmc/available_frequencies
```

```
root@topeet:/userdata# cat /sys/class/devfreq/dmc/available_frequencies
528000000 1068000000 1560000000 1596000000
root@topeet:/userdata#
```

设置频率

代码块

```
1 echo userspace | sudo tee /sys/class/devfreq/dmc/governor > /dev/null
2 echo 1596000000 | sudo tee /sys/class/devfreq/dmc/userspace/set_freq > /dev/null
```

查看

代码块

```
1 cat /sys/kernel/debug/clk/clk_summary | grep ddr
```

5. 快速定频

官方脚本: https://github.com/airockchip/rknn_model_zoo/blob/main/scaling_frequency.sh

代码块

```
1
2 chip_name=0
3 freq_set_status=0
4
5 usage()
6 {
7     echo "USAGE: ./scaling_frequency.sh -c {chip_name} [-h]"
8     echo "  -c:  chip_name, such as rv1126 / rk3588"
9     echo "  -h:  Help"
10 }
11
12 # print_and_compare_result want result
13 print_and_compare_result()
14 {
15     # echo "compare result"
16     echo "    try set \"$1"
17     echo "    query   \"$2"
18     if [ $1 == $2 ];then
19         echo "    Seting Success"
20     else
21         echo "    Seting Failed"
22         freq_set_status=-1
23     fi
24 }
25
26 # print_not_support_adjust {device} {freq want} {freq want}
27 print_not_support_adjust()
28 {
29     echo "Firmware seems not support seting $1 frequency"
30     echo "    wanted \"$2"
31     echo "    check  \"$3"
32 }
33
34
35 # MAIN FUNCTION HERE #
36 vaild=0
37
38 if [ $# == 0 ]; then
39     usage
40     exit 0
41 fi
42
43 while getopts "c:dc" arg
```

```
44  do
45      case $arg in
46          c)
47              chip_name=$OPTARG
48              ;;
49          h)
50              usage
51              exit 0
52              ;;
53          ?)
54              usage
55              exit 1
56              ;;
57      esac
58  done
59
60
61  if [ $chip_name == 'rv1126' ]; then
62      seting_strategy=1
63      CPU_freq=1512000
64      NPU_freq=934000000
65      DDR_freq=924000000
66  elif [ $chip_name == 'rv1109' ]; then
67      seting_strategy=1
68      CPU_freq=1512000
69      NPU_freq=594000000
70      DDR_freq=924000000
71  elif [ $chip_name == 'rk1808' ]; then
72      seting_strategy=1
73      # CPU_freq=1512000
74      CPU_freq=1200000
75      NPU_freq=792000000
76      DDR_freq=924000000
77  elif [ $chip_name == 'rk3399pro' ]; then
78      seting_strategy=2
79      CPU_freq=1512000
80      NPU_freq=792000000
81      DDR_freq=934000000
82  elif [ $chip_name == 'rk3562' ]; then
83      seting_strategy=6
84      CPU_freq=2000000
85      NPU_freq=1000000000
86      DDR_freq=1560000000
87  elif [ $chip_name == 'rk3566' ]; then
88      seting_strategy=3
89      CPU_freq=1800000
90      NPU_freq=1000000000
```

```
91     DDR_freq=1056000000
92     elif [ $chip_name == 'rk3568' ]; then
93         seting_strategy=3
94         CPU_freq=1800000
95         NPU_freq=1000000000
96         DDR_freq=1560000000
97     elif [ $chip_name == 'rk3588' ]; then
98         seting_strategy=4
99         CPU_freq=2256000
100        NPU_freq=1000000000
101        DDR_freq=2112000000
102    elif [ $chip_name == 'rk3576' ]; then
103        seting_strategy=7
104        CPU_freq=2016000
105        NPU_freq=1000000000
106        DDR_freq=2112000000
107    elif [ $chip_name == 'rv1106' ]; then
108        seting_strategy=5
109        CPU_freq=1608000
110        NPU_freq=500000000
111        DDR_freq=924000000
112    elif [ $chip_name == 'rv1103' ]; then
113        seting_strategy=5
114        CPU_freq=1608000
115        NPU_freq=500000000
116        DDR_freq=924000000
117    elif [ $chip_name == 'rv1126b' ]; then
118        seting_strategy=8
119        CPU_freq=1608000
120        NPU_freq=950000000
121        DDR_freq=100000000
122    else
123        echo "$chip_name not recognize"
124        exit 1
125    fi
126
127
128
129    echo "Try seting frequency for "${chip_name}
130    echo "    Setting strategy as $seting_strategy"
131    echo "    NPU seting to "$NPU_freq
132    echo "    CPU seting to "$CPU_freq
133    echo "    DDR seting to "$DDR_freq
134
135
136    case $seting_strategy in
137        0)
```

```

138     echo "seting strategy not implement now"
139     ;;
140
141     # for rk1808, rv1126, rv1109
142     1)
143         echo "CPU: seting frequency"
144         echo userspace > /sys/devices/system/cpu/cpu0/cpufreq/scaling_governor
145         echo $CPU_freq > /sys/devices/system/cpu/cpu0/cpufreq/scaling_setspeed
146         CPU_cur_freq=$(cat
/sys/devices/system/cpu/cpu0/cpufreq/cpuinfo_cur_freq)
147         print_and_compare_result $CPU_freq $CPU_cur_freq
148
149         echo "NPU: seting frequency"
150         echo userspace > /sys/class/devfreq/ffbc0000.npu/governor
151         echo $NPU_freq > /sys/class/devfreq/ffbc0000.npu/userspace/set_freq
152         NPU_cur_freq=$(cat /sys/class/devfreq/ffbc0000.npu/cur_freq)
153         print_and_compare_result $NPU_freq $NPU_cur_freq
154
155         echo "DDR: seting frequency"
156         if [ $chip_name == 'rk1808' ]; then
157             cur_freq=$(cat /sys/kernel/debug/clk/clk_summary | grep pll_dpll |
awk '{split($0,a," "); print a[4]}')
158         else
159             cur_freq=$(cat /sys/kernel/debug/clk/clk_summary | grep pll_dpll |
awk '{split($0,a," "); print a[5]}')
160         fi
161         DDR_cur_freq="`expr $cur_freq \* 2`"
162         print_not_support_adjust DDR $DDR_freq $DDR_cur_freq
163         ;;
164
165     # for rk3399pro
166     2)
167         echo "seting strategy not implement for rk3399pro now"
168         ;;
169
170     # for rk3566, rk3568
171     3)
172         echo 1 > /sys/devices/system/cpu/cpu0/cpuidle/state1/disable
173         echo 1 > /sys/devices/system/cpu/cpu1/cpuidle/state1/disable
174         echo 1 > /sys/devices/system/cpu/cpu2/cpuidle/state1/disable
175         echo 1 > /sys/devices/system/cpu/cpu3/cpuidle/state1/disable
176
177         echo "CPU: seting frequency"
178         echo userspace >
/sys/devices/system/cpu/cpufreq/policy0/scaling_governor
179         echo $CPU_freq >
/sys/devices/system/cpu/cpufreq/policy0/scaling_setspeed

```



```

180     CPU_cur_freq=$(cat
/sys/devices/system/cpu/cpu0/cpufreq/cpuinfo_cur_freq)
181     print_and_compare_result $CPU_freq $CPU_cur_freq
182
183     echo "NPU: seting frequency"
184     echo userspace > /sys/class/devfreq/fde40000.npu/governor
185     echo $NPU_freq > /sys/class/devfreq/fde40000.npu/userspace/set_freq
186     NPU_cur_freq=$(cat
/sys/devices/platform/fde40000.npu/devfreq/fde40000.npu/cur_freq)
187     print_and_compare_result $NPU_freq $NPU_cur_freq
188
189     echo "DDR: seting frequency"
190     echo userspace > /sys/class/devfreq/dmc/governor
191     echo $DDR_freq > /sys/class/devfreq/dmc/userspace/set_freq
192     DDR_cur_freq=$(cat /sys/class/devfreq/dmc/cur_freq)
193     print_and_compare_result $DDR_freq $DDR_cur_freq
194     ;;
195
196     # for rk3588
197     4)
198         echo 1 > /sys/devices/system/cpu/cpu0/cpuidle/state1/disable
199         echo 1 > /sys/devices/system/cpu/cpu1/cpuidle/state1/disable
200         echo 1 > /sys/devices/system/cpu/cpu2/cpuidle/state1/disable
201         echo 1 > /sys/devices/system/cpu/cpu3/cpuidle/state1/disable
202         echo 1 > /sys/devices/system/cpu/cpu4/cpuidle/state1/disable
203         echo 1 > /sys/devices/system/cpu/cpu5/cpuidle/state1/disable
204         echo 1 > /sys/devices/system/cpu/cpu6/cpuidle/state1/disable
205         echo 1 > /sys/devices/system/cpu/cpu7/cpuidle/state1/disable
206
207         echo "CPU: seting frequency"
208         echo "  Core0"
209         echo userspace >
/sys/devices/system/cpu/cpufreq/policy0/scaling_governor
210         echo 1800000 > /sys/devices/system/cpu/cpufreq/policy0/scaling_setspeed
211         CPU_cur_freq=$(cat
/sys/devices/system/cpu/cpu0/cpufreq/cpuinfo_cur_freq)
212         print_and_compare_result 1800000 $CPU_cur_freq
213
214         echo "  Core4"
215         echo userspace >
/sys/devices/system/cpu/cpufreq/policy4/scaling_governor
216         echo $CPU_freq >
/sys/devices/system/cpu/cpufreq/policy4/scaling_setspeed
217         CPU_cur_freq=$(cat
/sys/devices/system/cpu/cpu4/cpufreq/cpuinfo_cur_freq)
218         print_and_compare_result $CPU_freq $CPU_cur_freq
219

```

```

220     echo "   Core6"
221     echo userspace >
        /sys/devices/system/cpu/cpufreq/policy6/scaling_governor
222     echo $CPU_freq >
        /sys/devices/system/cpu/cpufreq/policy6/scaling_setspeed
223     CPU_cur_freq=$(cat
        /sys/devices/system/cpu/cpu6/cpufreq/cpuinfo_cur_freq)
224     print_and_compare_result $CPU_freq $CPU_cur_freq
225
226     echo "NPU: seting frequency"
227     if [ -e /sys/class/devfreq/fdab0000.npu/governor ];then
228         echo userspace > /sys/class/devfreq/fdab0000.npu/governor
229         echo $NPU_freq >
        /sys/class/devfreq/fdab0000.npu/userspace/set_freq
230         NPU_cur_freq=$(cat /sys/class/devfreq/fdab0000.npu/cur_freq)
231     elif [ -e /sys/class/devfreq/devfreq0/governor ];then
232         echo performance > /sys/class/devfreq/devfreq0/governor
233         NPU_cur_freq=$(cat /sys/class/devfreq/devfreq0/cur_freq)
234     else
235         NPU_cur_freq=$(cat /sys/kernel/debug/clk/scmi_clk_npu/clk_rate)
236     fi
237     print_not_support_adjust NPU $NPU_freq $NPU_cur_freq
238
239     echo "DDR: seting frequency"
240     if [ -e /sys/class/devfreq/dmc/governor ];then
241         echo userspace > /sys/class/devfreq/dmc/governor
242         echo $DDR_freq > /sys/class/devfreq/dmc/userspace/set_freq
243         DDR_cur_freq=$(cat /sys/class/devfreq/dmc/cur_freq)
244         print_and_compare_result $DDR_freq $DDR_cur_freq
245     else
246         print_not_support_adjust DDR $DDR_freq
247         $DDR_cur_freq=$(cat /sys/kernel/debug/clk/clk_summary | grep
scmi_clk_ddr)
248         echo $DDR_cur_freq
249     fi
250
251     ;;
252
253     # rv1106, rv1103
254     5)
255         echo "CPU: seting frequency"
256         echo userspace >
        /sys/devices/system/cpu/cpufreq/policy0/scaling_governor
257         echo $CPU_freq >
        /sys/devices/system/cpu/cpufreq/policy0/scaling_setspeed
258         CPU_cur_freq=$(cat
        /sys/devices/system/cpu/cpu0/cpufreq/cpuinfo_cur_freq)

```

```

259     print_and_compare_result $CPU_freq $CPU_cur_freq
260
261     echo "NPU: seting frequency"
262     # echo " no strategy to seting DDR frequency"
263     # echo $NPU_freq > /sys/kernel/debug/clk/aclk_npu_root/clk_rate
264     echo $NPU_freq > /sys/kernel/debug/clk/clk_500m_src/clk_rate
265     NPU_cur_freq=$(cat /sys/kernel/debug/clk/clk_summary | grep
aclk_npu_root | awk '{split($0,a," "); print a[5]}')
266     print_and_compare_result $NPU_freq $NPU_cur_freq
267
268     echo "DDR: seting frequency"
269     echo " no strategy to seting DDR frequency"
270     cur_freq=$(cat /sys/kernel/debug/clk/clk_summary | grep
clk_core_ddrc_src | awk '{split($0,a," "); print a[5]}')
271     DDR_cur_freq=`expr $cur_freq \* 2`
272     print_and_compare_result $DDR_freq $DDR_cur_freq
273     ;;
274
275     # rk3562
276     6)
277         echo 1 > /sys/devices/system/cpu/cpu0/cpuidle/state1/disable
278         echo 1 > /sys/devices/system/cpu/cpu1/cpuidle/state1/disable
279         echo 1 > /sys/devices/system/cpu/cpu2/cpuidle/state1/disable
280         echo 1 > /sys/devices/system/cpu/cpu3/cpuidle/state1/disable
281
282         echo "CPU: seting frequency"
283         echo userspace >
/sys/devices/system/cpu/cpufreq/policy0/scaling_governor
284         echo $CPU_freq >
/sys/devices/system/cpu/cpufreq/policy0/scaling_setspeed
285         CPU_cur_freq=$(cat
/sys/devices/system/cpu/cpu0/cpufreq/cpuinfo_cur_freq)
286         print_and_compare_result $CPU_freq $CPU_cur_freq
287
288         echo "NPU: seting frequency"
289         echo userspace >
/sys/devices/platform/ff300000.npu/devfreq/ff300000.npu/governor
290         echo $NPU_freq >
/sys/devices/platform/ff300000.npu/devfreq/ff300000.npu/userspace/set_freq
291         NPU_cur_freq=$(cat /sys/class/devfreq/ff300000.npu/cur_freq)
292         print_and_compare_result $NPU_freq $NPU_cur_freq
293
294         echo "DDR: seting frequency"
295         if [ -e /sys/class/devfreq/dmc/governor ];then
296             echo userspace > /sys/class/devfreq/dmc/governor
297             echo $DDR_freq > /sys/class/devfreq/dmc/userspace/set_freq
298             DDR_cur_freq=$(cat /sys/class/devfreq/dmc/cur_freq)

```

```

299         else
300             DDR_cur_freq=$(cat /sys/kernel/debug/clk/clk_summary | grep
scmi_clk_dds | awk '{split($0,a," "); print a[5]}')
301         fi
302         print_not_support_adjust DDR $DDR_freq $DDR_cur_freq
303     ;;
304
305     # rk3576
306     7)
307         echo 1 > /sys/devices/system/cpu/cpu0/cpuidle/state1/disable
308         echo 1 > /sys/devices/system/cpu/cpu1/cpuidle/state1/disable
309         echo 1 > /sys/devices/system/cpu/cpu2/cpuidle/state1/disable
310         echo 1 > /sys/devices/system/cpu/cpu3/cpuidle/state1/disable
311         echo 1 > /sys/devices/system/cpu/cpu4/cpuidle/state1/disable
312         echo 1 > /sys/devices/system/cpu/cpu5/cpuidle/state1/disable
313         echo 1 > /sys/devices/system/cpu/cpu6/cpuidle/state1/disable
314         echo 1 > /sys/devices/system/cpu/cpu7/cpuidle/state1/disable
315
316         echo "CPU: seting frequency"
317         echo "  Core0"
318         echo userspace >
/sys/devices/system/cpu/cpufreq/policy0/scaling_governor
319         echo 1800000 > /sys/devices/system/cpu/cpufreq/policy0/scaling_setspeed
320         CPU_cur_freq=$(cat
/sys/devices/system/cpu/cpufreq/policy0/scaling_cur_freq)
321         print_and_compare_result 1800000 $CPU_cur_freq
322
323         echo "  Core4"
324         echo userspace >
/sys/devices/system/cpu/cpufreq/policy4/scaling_governor
325         echo $CPU_freq >
/sys/devices/system/cpu/cpufreq/policy4/scaling_setspeed
326         CPU_cur_freq=$(cat
/sys/devices/system/cpu/cpufreq/policy4/scaling_cur_freq)
327         print_and_compare_result $CPU_freq $CPU_cur_freq
328
329         echo "NPU: seting frequency"
330         echo userspace >
/sys/devices/platform/27700000.npu/devfreq/27700000.npu/governor
331         echo $NPU_freq > /sys/class/devfreq/27700000.npu/userspace/set_freq
332         NPU_cur_freq=$(cat /sys/class/devfreq/27700000.npu/cur_freq)
333         print_and_compare_result $NPU_freq $NPU_cur_freq
334
335         echo "DDR: seting frequency"
336         if [ -e /sys/class/devfreq/dmc/governor ];then
337             echo userspace > /sys/class/devfreq/dmc/governor
338             echo $DDR_freq > /sys/class/devfreq/dmc/userspace/set_freq

```

```

339         DDR_cur_freq=$(cat /sys/class/devfreq/dmc/cur_freq)
340     else
341         DDR_cur_freq=$(cat /sys/kernel/debug/clk/clk_summary | grep
scmi_clk_ddr | awk '{split($0,a," "); print a[5]}')
342     fi
343     print_not_support_adjust DDR $DDR_freq $DDR_cur_freq
344     ;;
345
346     # rv1126b
347     8)
348         echo "CPU: seting frequency"
349         # cat
/sys/devices/system/cpu/cpu0/cpufreq/scaling_available_frequencies
350         echo userspace > /sys/devices/system/cpu/cpu0/cpufreq/scaling_governor
351         echo $CPU_freq > /sys/devices/system/cpu/cpu0/cpufreq/scaling_setspeed
352         CPU_cur_freq=$(cat
/sys/devices/system/cpu/cpufreq/policy0/scaling_cur_freq)
353         print_and_compare_result $CPU_freq $CPU_cur_freq
354
355         echo "NPU: seting frequency"
356         # # cat
/sys/devices/platform/220000000.npu/devfreq/220000000.npu/available_frequencies
357         echo userspace >
/sys/devices/platform/220000000.npu/devfreq/220000000.npu/governor
358         echo $NPU_freq > /sys/class/devfreq/220000000.npu/userspace/set_freq &
359         NPU_cur_freq=$(cat /sys/class/devfreq/220000000.npu/cur_freq)
360         print_and_compare_result $NPU_freq $NPU_cur_freq
361
362         echo "DDR: seting frequency"
363         if [ -e /sys/class/devfreq/dmc/governor ];then
364             echo userspace > /sys/class/devfreq/dmc/governor
365             echo $DDR_freq > /sys/class/devfreq/dmc/userspace/set_freq
366             DDR_cur_freq=$(cat /sys/class/devfreq/dmc/cur_freq)
367         else
368             DDR_cur_freq=$(cat /sys/kernel/debug/clk/clk_summary | grep
pclk_ddrc | awk '{split($0,a," "); print a[5]}')
369         fi
370         print_not_support_adjust DDR $DDR_freq $DDR_cur_freq
371         ;;
372
373     *)
374         echo "seting strategy not implement now"
375         ;;
376 esac
377
378
379     echo "freq set" > ./freq_set_status

```

```
380 echo "CPU: "$CPU_freq >> ./freq_set_status
381 echo "NPU: "$NPU_freq >> ./freq_set_status
382 echo "DDR: "$DDR_freq >> ./freq_set_status
383 echo "freq query" >> ./freq_set_status
384 echo "CPU: "$CPU_cur_freq >> ./freq_set_status
385 echo "NPU: "$NPU_cur_freq >> ./freq_set_status
386 echo "DDR: "$DDR_cur_freq >> ./freq_set_status
387 if [ $freq_set_status == 0 ];then
388     echo "Seting Success" >> ./freq_set_status
389 else
390     echo "Seting Failed" >> ./freq_set_status
391 fi
```